

Last name and first name:

Politecnico di Milano

Masters of Science in Computer Science and Engineering; Geoinformatics Engineering;
Music and Acoustic Engineering; and Telecommunication Engineering

FOUNDATIONS OF OPERATIONS RESEARCH

Exam of January 12, 2022

M.Sc. in:

Last name:

First name:

Student number:

Prob. 1	Prob. 2	Prob. 3	Prob. 4	Prob. 5	Total
8.0	8.5	7.0	3.0	4.5	31

Time available: 2 hours and 15 minutes

Write the answers to the various questions in the specified locations.

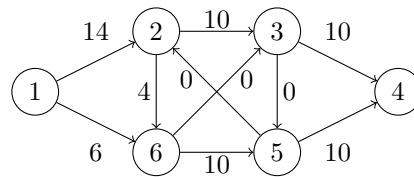
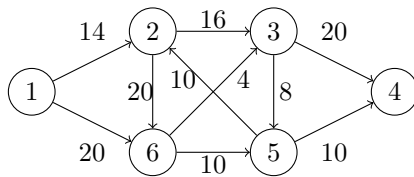
Report and explain all steps on the back of the previous page.

The answers should be as clear and precise as possible.

No course material or summary is allowed, except for Problem 5 which is open notes/book.

Problem 1

Let $G = (N, A)$ be the graph depicted on the left, with 1 and 4 as the source and the sink nodes, respectively:



where the value associated to each arc (i, j) represents the arc capacity, k_{ij} .

a) Apply two iterations of the Ford-Fulkerson algorithm to the maximum flow problem from node 1 to node 4 in graph G , starting from the feasible flow in the graph on the right.

Indicate here the resulting feasible flow (specify the amount of product crossing each arc and the overall flow value) and report all the steps on the back of the previous page.

b) Consider the graph G and replace now k_{34} with 10. Show that the flow in the picture on the right is optimal and determine a cut separating node 1 from node 4 of minimum capacity in graph G .

c) Assume that the value associated with each arc in the original graph G represents its cost. Describe an algorithm studied in the course that can solve the shortest path problem from node 1 to all the other nodes in G . Mention its computational complexity in terms of the number of nodes n and the number of arcs m of G and motivate your answer.

Problem 2

Consider the linear programming problem:

$$\begin{array}{llllll} \max & -6x_1 & -8x_2 & +6x_3 & & \\ \text{s. t.} & -x_1 & -2x_2 & -x_3 & \leq & 1 \\ & 3x_1 & -4x_2 & +3x_3 & \leq & 1 \\ & x_1, & x_2, & x_3 & \geq & 0 \end{array}$$

a) Write the problem in the standard form and indicate a basic solution associated with it. Comment on the feasibility of this solution.

b) Apply one iteration of the simplex method with Bland's rule to this linear program, starting from the solution $\underline{x} = (0, 0, 0)^T$. Report all the steps, including the initial tableau and the resulting tableau. Check whether the obtained solution is optimal and motivate your answer.

c) Write the dual of the original problem.

d) Derive the optimal solution of the dual problem from the optimal solution $\underline{x}^* = (0, 0, 1/3)^T$ of the given linear program. Clearly state the result(s) that justifies(y) your answer.

Problem 3

Consider the following Integer Linear Programming (ILP) problem:

$$\begin{array}{llll} \max & -x_1 & +3x_2 & \\ \text{s. t.} & 2x_1 & +2x_2 & \leq 7 \\ & -4x_1 & +4x_2 & \leq 6 \\ & x_1, & x_2 & \geq 0 \text{ integer} \end{array}$$

a) Find the optimal solution of the above ILP problem with the Branch-and-Bound method by solving graphically the linear programming relaxation of the subproblems.

Report all the steps (branching tree, the graphical solutions of all subproblems with the corresponding optimal solutions and objective function values) on the back of the previous page.

Indicate here the optimal solution found and its objective function value.

b) Provide the definition of a cutting plane for a general ILP problem.

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Problem 4

Write below the AMPL .mod file for the following variant of the maximum flow problem:

- a set V : nodes
- a set $A \subseteq V \times V$: arcs
- a set $B \subseteq A$
- parameters $s, t \subseteq V$, respectively the source and the sink nodes
- a non negative parameter f_{ij} for all $(i, j) \in A$
- a non negative variable ϕ
- a non negative variable x_{ij} for all $(i, j) \in A$
- the following objective function to be maximized: $\max \phi$
- the following constraints:

$$\sum_{(i,j) \in A} x_{ij} - \sum_{(j,i) \in A} x_{ji} = \begin{cases} \phi, & i = s \\ -\phi, & i = t \\ 0 & i \in V, i \neq s, i \neq t \end{cases}$$
$$x_{ij} \leq f_{ij}, (i, j) \in A \setminus B$$

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Problem 5

Consider a computer system with a set $I = \{1, 2, \dots, n\}$ of users and a set $J = \{1, 2, \dots, m\}$ of possible hub locations. Let c_{ij} denote the cost for connecting user i to the hub located at point j , for $i \in I, j \in J$; d_j denote the capacity unitary cost for using the hub located at j , for $j \in J$; and v_j denote the fixed cost of establishing a hub at point j , for $j \in J$.

Limitations on the hubs, such as number of input ports, available memory space, address structure and resource sharing schemes, constrain the subset of users that can be connected to each hub. It is assumed that a_{ij} is the capacity required to a hub located at j to serve user i , if user i is connected to hub j , for $i \in I, j \in J$, and that b_j is the maximum capacity of the hub that can be installed at location j , for $j \in J$.

Let $I' \subset I$ denote a subset of k users requesting premium service with $k \geq 3$. Then, for technical reasons, no more than $k/2$ users of I' can be connected to the same hub.

The problem is to determine how to locate the hubs and assign the users to the hubs, so as to minimize the total costs while respecting the various constraints.

Give an integer linear programming formulation for the problem. For the sake of simplicity, assume that there exists feasible solutions for the considered parameter values.

Decision variables:

Objective function:

Constraints: